

IMPERIAL COUNTY AP, CA, USA

WMO: 747185

Lat: 32.834N

Lon: 115.579W

Elev: -18

StdP: 101.54

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 03144

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.2	3.6	-12.7	1.3	16.1	-10.5	1.5	16.2	12.0	17.2	10.7	17.5	2.1	260	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.6	44.4	22.6	43.0	22.4	42.1	22.2	27.2	36.2	26.4	35.7	25.6	35.5	3.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.0	31.4	23.9	18.7	31.6	22.7	17.4	31.8	85.2	36.1	81.8	35.4	78.7	35.6	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	10.0	8.4	-0.7	47.3	1.9	1.6	-2.1	48.5	-3.2	49.4	-4.3	50.3	-5.7	51.5
			-3.4	28.8	1.7	1.5	-4.6	29.8	-5.6	30.7	-6.5	31.6	-7.7	32.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	13.8	15.3	19.0	21.8	25.9	30.8	34.2	34.3	30.9	24.1	17.9	13.1
	DBStd	8.10	2.78	3.25	3.44	3.50	3.62	3.09	2.05	2.08	3.14	3.55	3.42	2.84
	HDD10.0	13	4	2	0	0	0	0	0	0	0	0	1	6
	HDD18.3	496	142	93	35	10	1	0	0	0	0	3	49	163
	CDD10.0	4927	122	150	278	353	493	625	750	753	627	437	237	102
	CDD18.3	2368	1	8	54	113	236	375	492	495	377	182	35	1
	CDH23.3	36148	60	188	825	1521	3222	5735	8072	8100	5648	2254	489	34
CDH26.7	23312	3	38	327	736	1918	3854	5688	5698	3690	1202	155	2	
Wind	WSAvg	3.2	2.4	2.9	3.5	4.2	4.1	3.9	3.5	3.1	2.9	2.8	2.5	2.3
Precipitation	PrecAvg	61	11	7	6	2	0	0	2	3	8	5	8	7
	PrecMax	125	38	34	25	16	2	0	29	12	76	47	44	41
	PrecMin	18	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	31	13	9	7	5	0	0	6	4	18	11	13	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	30.0	34.4	37.6	42.2	45.5	46.3	45.9	43.5	38.9	32.5	26.3
		MCWB	13.7	14.8	16.3	17.9	19.8	21.2	23.1	23.1	22.0	20.0	15.9	14.3
	2%	DB	25.1	27.8	32.3	35.2	39.2	43.3	44.5	44.2	42.1	37.0	30.4	24.0
		MCWB	12.8	14.2	15.8	16.9	18.8	20.9	22.9	23.0	21.8	19.4	15.3	12.9
	5%	DB	23.3	25.9	30.3	33.1	37.6	41.9	42.9	42.8	40.8	35.0	28.5	22.4
		MCWB	12.2	13.8	15.2	16.2	18.9	20.4	22.6	23.2	21.6	18.0	15.0	12.1
	10%	DB	21.8	23.5	28.1	31.1	35.8	40.2	41.9	41.7	39.0	33.0	26.3	20.5
		MCWB	11.9	12.8	14.6	15.6	17.9	20.0	22.6	23.2	21.7	17.5	14.3	11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	17.6	18.6	19.9	23.0	25.7	27.7	28.9	27.7	23.9	18.3	16.3
		MCDB	23.5	25.9	29.5	32.7	35.9	38.6	38.6	37.7	35.8	32.8	27.1	22.7
	2%	WB	15.0	15.9	16.9	18.3	21.0	24.3	26.7	27.4	26.4	21.9	16.8	14.8
		MCDB	21.4	23.9	29.1	31.9	34.5	36.5	36.9	35.8	34.7	30.5	26.9	21.0
	5%	WB	13.8	14.6	16.0	17.3	19.8	23.0	25.9	26.7	25.3	20.3	15.8	13.6
		MCDB	20.5	23.1	28.1	30.8	34.5	35.9	36.2	35.5	34.5	30.7	26.4	19.7
	10%	WB	12.6	13.6	15.0	16.2	18.7	21.8	25.2	25.9	24.3	18.8	14.9	12.3
		MCDB	19.7	21.9	26.7	29.5	33.4	35.8	35.8	35.3	34.2	30.8	25.2	18.6
Mean Daily Temperature Range	5% DB	MDBR	15.0	14.7	15.8	16.1	16.7	17.5	15.1	14.6	15.4	16.0	15.4	14.4
		MCDBR	17.6	17.7	18.6	18.7	19.0	19.2	16.7	16.1	16.7	18.0	17.6	16.5
		MCWBR	9.4	8.6	8.0	7.5	7.0	7.1	6.2	6.3	6.4	7.3	8.2	8.8
	5% WB	MCDBR	14.1	15.0	17.2	17.4	18.1	17.2	13.4	13.0	13.2	14.8	15.6	13.5
		MCWBR	8.4	8.1	8.0	7.4	7.2	7.2	5.8	5.8	7.0	7.0	7.8	7.6
Clear-Sky Solar Irradiance	taub		0.291	0.295	0.313	0.330	0.348	0.349	0.393	0.388	0.351	0.321	0.295	0.291
	taud		2.514	2.523	2.457	2.411	2.379	2.380	2.365	2.372	2.437	2.490	2.531	2.540
	Ebn at Noon		936	967	967	958	937	930	888	890	914	923	924	917
	Edh at Noon		84	92	107	117	122	122	123	120	108	94	82	78
All-Sky Solar Radiation	RadAvg		3.50	4.47	5.93	7.25	8.08	8.43	7.66	7.12	6.28	5.05	3.83	3.16
	RadStd		0.18	0.31	0.21	0.18	0.22	0.29	0.31	0.28	0.23	0.20	0.17	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (23 neighbors)	+0.52	N/A	-1.58	N/A	N/A	N/A	N/A	N/A	+217	+163
	+0.48	N/A	-1.29	+0.59	N/A	N/A	N/A	-76	+174	+91

Nomenclature: See separate page